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ASSESSMENT, DEVELOPMENT, AND VALIDATION

Psychometric Analysis of the Survey of Perceived Organizational Support (SPOS-8) Using 1,005 School Counselors

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ABSTRACT
Psychometric properties of the Survey of Perceived Organizational Support (SPOS-8) scores were examined with a national sample of school counselors ($N = 1{,}005$). SPOS-8 scores displayed good internal consistency (α = .92), one-factor structure, and convergent and discriminant validity. Measurement invariance was evident for White and racial/ethnic minority school counselors.

KEYWORDS
Survey of perceived organizational support; SPOS-8; organizational support; psychometric analysis; school counselor

Professionals in high work demand occupations that require sustained emotional commitment are especially vulnerable to negative job-related outcomes, such as exhaustion and burnout (Griffin et al., 2010). Researchers examining employee wellbeing have focused on the role of job resources that buffer the harmful effects of work stress and enhance job satisfaction (Schaufeli & Taris, 2014). Perceived organizational support, defined as employees' sense of support when organizations acknowledge their contributions and pay attention to employee well-being (Eisenberger et al., 1986) has been widely identified as an essential job resource not only for employee retention, but also for promoting employee engagement, commitment, and professional development (Brunetto et al., 2013; Pati & Kumar, 2010; Rhoades et al., 2001; Riggle et al., 2009). Furthermore, support from supervisors and coworkers serves as a crucial source of organizational social support that can offset work stressors stemming from high demands and enhance coping skills in the workplace (Gray-Stanley & Muramatsu, 2011).

The Survey of Perceived Organizational Support (SPOS) is one of the most frequently used scales for assessing employee perceptions of how the organization provides employee support and has been used in over 160 studies. Since its development more than 35 years ago (Eisenberger et al., 1986), researchers have used the SPOS to examine this construct across emotionally demanding occupations, such as nursing, teaching, human services, and counseling. In Web of Science, the Eisenberger et al. (1986) article introducing the SPOS has been cited more than 3,400 times. In addition to the original 36-item scale, researchers frequently selected several shorter versions of the SPOS to measure this construct in a wide range of studies examining the linkages between work environments and related employee outcomes. Indeed, due to practical considerations of ease and efficiency, researchers employed the shorter versions of the SPOS more frequently than the full scale (Rhoades & Eisenberger, 2002), with the 8-item short version (SPOS-8) being one of the most commonly administered short versions (Hutchison, 1997b). These eight items were selected from the original SPOS due to desirable factor loadings and conceptual alignment. Based on a 7-point Likert-type scale (1 = *totally disagree* to 7 = *totally agree*), the SPOS-8 scores indicated strong internal consistency over .90 (Hutchison, 1997b; Rhoades & Eisenberger, 2002). The unidimensional model of the SPOS-8 was supported by

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several studies (e.g., Rhoades & Eisenberger, 2002). The convergent and discriminant validity were identified by examining the correlations between the SPOS-8 scores and other robust instruments that assess organizational commitment, job satisfaction, and supervisory support, among others (Hutchison, 1997a, 1997b; Tourangeau et al., 2010).

Despite the robustness and widespread use of the SPOS-8, there is still a notable lack of psychometric evidence within specific professions, even though organizational support is a salient construct for professional performance across all human services professions. One notable example is school counseling. Existing school counseling literature has repeatedly identified the difficulties that school counselors encounter in the workplace, including demanding caseloads, burdensome non-counseling duties, and role conflict (Bardhoshi et al., 2014; Culbreth et al., 2005). Given this context, organizational support can serve as a critical buffer between job challenges and work-related outcomes (Ren & Zhang, 2015; Wallace et al., 2009), making this construct relevant and important for both school counseling research and practice. Moreover, previous findings suggested that perceptions and experiences of organizational support are influenced by each individual's race/ethnicity across various occupations (Avery et al., 2007; Payne et al., 2018), which necessitates the consideration of racial/ethnic minority status in understanding school counselors' perceived organizational support.

Existing literature indicates that for school counselors, perceptions of being valued, rewarded, and acknowledged by workplace leaders and supervisors lead to increased self-efficacy (Konczak et al., 2000). As indicated by Bardhoshi and Um (2021), self-efficacious beliefs are promoted through an encouraging school climate, which provides school counselors with opportunities to experience positive and supportive feedback. Furthermore, perceived organizational support shows a high association with school counselor job satisfaction (Bardhoshi et al., 2019). Given that a school counselor is also a part of the school environment, social and emotional support from the school has a reciprocal relationship with school counselors' commitment and wellness (Bernard & Goodyear, 2004). Additionally, as school administrators play an important role in the overall school counselor workplace experience, their support can facilitate school counselor accomplishment of counseling duties that lead to positive student outcomes (Randick et al., 2018). Thus, organizational support not only significantly influences school counselors' professional and individual wellness, it also has important implications for the wellbeing of their students. While we identified one study administering the 17-item SPOS scale to 142 school counselors from two school districts (see Geigel, 2013), there is a substantial need to understand the contribution of perceived organizational support to school counselor performance and wellness in larger, more diverse, national samples, using a more succinct, easy to administer version. Although the SPOS-8 has been widely used for assessing perceived support from workplaces in mental health professionals (Brunetto et al., 2013; Kim & Barak, 2015; Lizano & Barak, 2012), its adequacy with a school counselor population has not been tested. Given that school counselors practice in a very different setting compared to other mental health professionals, and that organizational support in a school setting encompasses multiple aspects that are critical for school counselors' work-related attitudes and outcomes (Geigel, 2013; Kim & Lambie, 2018; Wilkerson & Bellini, 2006), it is important to identify a measure that yields reliable and valid scores and assesses school counselors' perceived organizational support.

The purpose of this current study was to examine the psychometric properties of SPOS-8 scores from a nationwide sample of 1,005 professional school counselors. The specific research questions were as follows: (a) What are the descriptive statistics of the SPOS-8 for each item and total score with a national sample of practicing school counselors?; (b) What is the internal consistency of the SPOS-8 scores for the national sample of practicing school counselors?; (c) Does our sample data support the one-factor model of SPOS-8 as suggested in previous literature?; (d) Does the SPOS-8 indicate measurement invariance across race/ethnicity (i.e., White and racial/ethnic minority school counselors)?; and (e) Do the SPOS-8 scores significantly correlate with measures of school counselor burnout, self-efficacy, job satisfaction, and perceived school principal support? Since this is the first attempt to apply the SPOS-8 to a large national sample of school counselors, we expect the results can be a useful reference for exploring school counselors' perceived organizational support in future studies.

Method

The dataset used for this psychometric study was part of a larger effort investigating the effects of school counselors' job demands and resources on their professional experiences in K-12 settings, including burnout, self-efficacy, and job satisfaction (Bardhoshi & Um, 2021), as well as the robustness of frequently used instruments in measuring these constructs in school counselors. Bardhoshi and Erford (2022) have examined other psychometric instruments with this national dataset to comprehensively evaluate their performance with a large sample of school counselors, in an effort to identify relevant, accurate, and efficient tools that measure key school counselor organizational variables that impact their professional trajectories.

Participants

A total of 1,005 practicing school counselors were recruited from the American School Counseling Association (ASCA) professional membership database ($N = 9{,}381$). The participants completed an online survey administered through SurveyMonkey that included the Survey of Perceived Organizational Support (SPOS-8; Eisenberger et al., 1986), the Counselor Burnout Inventory (CBI; Lee et al., 2007), the School Counselor Self-Efficacy Scale (SCSE; Bodenhorn & Skaggs, 2005), a single-item Global Job Satisfaction measure (Scarpello & Campbell, 1983), a single item measuring school principal support, demographic questions, and seven open-ended qualitative questions regarding school counselor training and job responsibilities.

Among the participants, 870 (86.6%) self-identified as female, while 131 (13%) and 4 (0.4%) self-identified as male and transgender, respectively. According to the self-report of race/ethnicity, 841 (83.6%) were of European descent, 76 (7.6%) were of African descent, 44 (4.4%) were of Hispanic descent, 20 (2.1%) were multi-racial, 8 (0.8%) were of Native Hawaiian/Pacific Islander descent, 8 (0.8%) were of Asian descent, 5 (0.5%) were of American Indian/Alaskan Native decent, and 3 (0.3%) were of some other race/ethnicity. The participants were employed in diverse types of schools, including high-schools ($n = 363$, 36.1%), elementary schools ($n = 258$, 25.7%), middle schools ($n = 208$, 20.7%), K-12 schools ($n = 59$, 5.9%), 6–12 schools ($n = 33$, 3.3%), or others ($n = 84$, 8.3%). In terms of educational level, 842 (83.8%) had a master's degree, 104 (10.3%) had an Ed.S degree, and 55 (5.4%) had a Ph.D or Ed.D degree. Four participants (0.4%) had only a Bachelor's degree and were included in this study. Regarding school counseling experience, 432 (43%) of the school counselors had worked for 11 or more years, 336 (33.4%) had 6–10 years of experience, and 237 (23.6%) had 0–5 years of experience. Lastly, according to the categorization used by the American Counseling Association, participants indicated their regional representations as Midwest ($n = 428$, 42.6%), Southern ($n = 291$, 29%), Western ($n = 147$, 14.6%), or North Atlantic ($n = 139$, 13.8%).

Instruments

In addition to the SPOS-8, we administered the CBI, SCSE, a global job satisfaction question, and a school principal support question to the participants. Additional measurements were selected considering their conceptual linkages with perceived organizational support based on a review of previous literature. Details on each measure are included below.

Survey of Perceived Organization Support (SPOS-8)

The SPOS-8 is an 8-item short form of the Survey of Perceived Organizational Support (Eisenberger et al., 1986). Response to each item is based on seven Likert response options ranging from 1 = *totally disagree* to 7 = *totally agree*. Hutchison (1997b) verified structural validity by yielding an adequate fit of the one-factor model, as well as convergent validity by examining correlations with other measures including the Affective Commitment Scale and Survey of Perceived Supervisory Support, among others. To date, studies on the SPOS-8 supported strong internal reliability of scores as .90 and construct validity as a one-factor model (see Eisenberger et al., 1986; Rhoades & Eisenberger, 2002). In the current study, we examined the eight items and changed the word "organization" to "school" in order to measure each school counselor's perceived support from the school work setting. The observed value of α for the SPOS-8 scores from the current sample data was .92.

Counselor Burnout Inventory (CBI)

The CBI is a 20-item self-report measurement developed by Lee et al. (2007), designed to assess burnout of professional school counselors, and accompanied by five ordinal response options ranging from 1 = *never true* to 5 = *always true*. This scale contains five subscales: Exhaustion, Incompetence, Negative Work Environment, Devaluing Client, and Deterioration in Personal Life, including four items in each subscale. Therefore, the score of each subscale falls between 4 and 20, and the CBI total score falls between 20 and 100. Lee et al. (2007) reported strong reliability across subscales including internal consistency (α = .73–.85) and test-retest dependability coefficients with a 6-week retest interval (r = .72–.85). Convergent validity based on a five-factor model was also robust across multiple studies (see Bardhoshi et al., 2019). In this study, the coefficient alpha of the CBI was .90 for the total scale and ranged from .66 to .89 for the subscales. This scale is contained in our questionnaire because support from workplace settings (e.g., interpersonal relationship, appropriate rewards, available resources) is associated with burnout of employees including counselors (Jawahar et al., 2007; Thompson et al., 2014).

School Counselor Self-Efficacy Scale (SCSE)

The SCSE is a 43-item self-report scale consisting of five dimensions: Personal and Social Development (12 items), Leadership and Assessment (9 items), Career and Academic Development (7 items), Collaboration (11 items), and Cultural Acceptance (4 items). This scale is designed to assess self-efficacy of school counselors using five ordinal options ranging from 1 = *not confident* to 5 = *highly confident*. Bodenhorn and Skaggs (2005) reported high coefficient alphas for subscales, ranging from .72 to .91. The current study sample also revealed strong internal consistency of .96 for the total scale scores and .71 to .90 for the subscale scores. EFA results supported a five-factor model, explaining 55% of the total variance, although Bardhoshi and Erford (2022) determined four and five factor models best fit the data from a large sample of school counselors. Given the perspective that a perception of environmental support impacts employees' self-efficacious beliefs (Bardhoshi & Um, 2021), the SCSES was added in the present study.

Job Satisfaction

A single-item global measure of job satisfaction developed by Scarpello and Campbell (1983) was used to assess school counselor job satisfaction. The item, "Overall, how satisfied are you with your job?" was rated using five ordinal response options ranging from 1 = *not at all* to 5 = *very much so*. Existing literature about this overall job satisfaction item supported its validity and usefulness, compared to multiple-item job satisfaction measures (Wanous et al., 1997), and Nagy (2002) supported its potential as a viable option when practical reasons (e.g., cost, length) make the use of a single-item measure preferable for measuring overall job satisfaction. The linkage between job satisfaction and organizational support was robust across studies (e.g., Eisenberger et al., 1997), therefore we planned to apply it to school counselors.

School Principal Support

Based on the practical grounds for using a single-item global job satisfaction scale, a single item designed to measure satisfaction with school principal support was also included. The item, "Overall, how supported do you feel by your school principal?" was rated using five ordinal response options ranging from 1 = *not at all* to 5 = *very much so*. Since school administrators play an important role in forming positive work environments (see Halawah, 2005), a high correlation was expected between the SPOS-8 and satisfaction with school principal support.

Procedure

The authors employed a recruitment e-mail to every professional member of the ASCA online school counselor membership directory ($N = 9{,}381$). We obtained delivery confirmation for 9,266 e-mails, and 115 e-mails were returned as non-deliverable. We applied a three-wave contact procedure recommended by Dillman et al. (2014) for distribution of Internet surveys. Participants received a hyperlink to an online survey, which included a welcome screen with continuing instructions, informed consent, a criterion sampling page, and all measures. To ensure participant's eligibility (Mills & Gay, 2017), we asked participants to confirm their (a) status of active school counselor certification/licensure in their practicing state and (b) work settings that included elementary, middle, and/or high school before they could proceed with the rest of the survey. A total of 1,069 participants (11.5%) submitted the online survey. After excluding 64 participants who did not finish one or more full component, the final sample included 1,005 school counselors (10.5% of the original sample).

Analysis

A very small portion (less than 2% overall) of the data were missing, and these values were determined to be missing completely at random (MCAR) as indicated by Little's MCAR test and addressed using mean imputation because the data MCAR would not be biased depending on the imputation method (Sterner, 2011). Descriptive statistics, corrected item-total correlations, squared multiple correlations, Cronbach's alpha coefficient, and bivariate correlations between the SPOS-8 and other robust measures were calculated using SPSS 27. Single- and multi-group confirmatory factor analysis (CFA) were conducted by using Mplus 8.6 (Muthen & Muthen, 2019) to determine the fit of the data to the one-factor model generated in previous studies for structural validity (Hutchison, 1997b; Rhoades et al., 2001). For conducting multi-group CFA, we used the sample of 144 White school counselors and 144 racial/ethnic minority (REM) school counselors. Although there is not a unified consensus on the exact number of participants needed to adequately power such analysis, several researchers suggested that for multi-group CFA, over 100 participants in each group is considered a guideline (Finch et al., 2018; Kyriazos, 2018; Wang & Wang, 2012). Our sample size also satisfied the minimum N:p (subjects to variables) ratio of 5 to 1 (Brown, 2015; Kyriazos, 2018). To identify the model fit, we used several goodness-of-fit indices including the comparative fit index (CFI), Tucker-Lewis index (TLI), root-mean-square error of approximation (RMSEA), and standardized root-mean square residual (SRMR). According to Hooper et al. (2008), an adequate model fit includes a nonsignificant chi-square, CFI and TLI of .90 or greater, and RMSEA and SRMR of .08 or lower. If CFI and TLI is .95 or greater, the model fit is considered excellent. Measurement invariance differences between groups were assessed using Chen's (2007) guidelines of ΔCFI >.01 and ΔRMSEA >.015.

Results

Preliminary Analysis, Descriptive Statistics, and Internal Consistency

According to Little's missing completely at random (MCAR) test, our data were MCAR (χ^2 = 81.89, $df = 92$, p = .766). Therefore, mean imputation was used to address the missing values (less than 2% overall). Because univariate skewness (−1.92 to 1.08) and kurtosis (−1.15 to 4.22), and multivariate kurtosis did not satisfy the normality assumption, we used maximum likelihood estimation of parameters with robust standard errors (i.e., MLR), which is applicable to non-normal data. Descriptive statistics, including means, standard deviations, corrected item-total correlations,

Table 1. Descriptive Statistics and Bivariate Correlation Coefficients Between the SPOS-8 Items.

SPOS-8 Item	1	2	3	4	5	6	7	8
1. The school values my contribution to its well-being. (1)	–							
2. The school fails to appreciate any extra effort from me.[R] (3)	.47	–						
3. The school would ignore any complaint from me.[R] (6)	.52	.60	–					
4. The school really cares about my well-being. (9)	.57	.48	.51	–				
5. Even if I did the best job possible, the school would fail to notice.[R] (17)	.57	.68	.71	.53	–			
6. The school cares about my general satisfaction at work. (21)	.56	.49	.50	.61	.57	–		
7. The school shows very little concern for me.[R] (23)	.61	.63	.67	.62	.78	.63	–	
8. The school takes pride in my accomplishments at work. (27)	.63	.54	.55	.60	.62	.67	.64	–
M	6.03	4.88	5.53	5.60	5.45	5.10	5.58	5.75
SD	1.19	1.87	1.54	1.47	1.70	1.62	1.57	1.56
Corrected Item-Total Correlation	.68	.69	.71	.69	.80	.70	.82	.75
Squared Multiple Correlation	.50	.52	.55	.51	.70	.55	.69	.60

Note. $N = 1{,}005$; All 8 items have both theoretical and empirical ranges of 1-7. Each item number in the parentheses is from the original scale (Eisenberger et al., 1986). [R] indicates that the item scores were reverse-coded. All correlations were significant at $p < .05$.

and squared multiple correlations of each SPOS-8 item/item-pair are presented in Table 1 as a reference for future school counseling research. Overall, mean scores from each item and total scale showed a high level of school counselors' perceptions about organizational support from 4.88 to 6.02 (1 = *totally disagree*; 7 = *totally agree*). The coefficient alpha for the SPOS-8 scores was .92, indicating a robust internal consistency for an 8-item measure.

Results of Confirmatory Factor Analysis (CFA)

Baseline Model

For confirming the baseline model of this study, we conducted CFA for the participants ($n = 1{,}005$). The result of CFA using MLR for continuous variables supported a one-factor model, yielding the fit indices of: CFI = .927; TLI = .898; RMSEA = .097 [90% CI: .085–.109]; and SRMR = .038. Chi-square for the analysis was 209.52 with 20 degrees of freedom ($p < .001$), although researchers recommend the utilization of alternative fit indices instead of the chi-square statistics due to less sensitivity to violation of the normality assumption and sample size (Chen, 2007; Kyriazos, 2018). The absolute values of standardized factor loadings ranged from .71 to .87. Additional information about the internal structure within the SPOS-8 was not investigated due to the nonexistence of subscales. Given the inadequate RMSEA value, we included three residual covariances following these criteria: (a) the residual covariance improved the model fit across different samples of this study; (b) the modification indices (MIs) of the residual covariance were not too substantial, in order to avoid the chances of sampling error and considerable changes in estimates; and (c) the correlated items shared similar contents in the same factor (Ghosh et al., 2021). Finally, the residual covariances were set between Item 1 and Item 8, Item 4 and Item 5, and Item 6 and Item 8. As a consequence of conducting CFA using MLR for the modified one-factor model, the model fit indices indicated excellent fit as shown in Table 2: $x^2(17) = 86.57$ ($p < .001$); CFI = .973; TLI = .956; RMSEA = .064 [90% CI: .044–.087]; and SRMR = .027, with the absolute values of standardized factor loadings ranging from .69 to .89.

Measurement Invariance Testing

In order to test the identified baseline model (i.e., modified one-factor model) for White/REM school counselors, we performed single-group CFAs. We excluded 20 participants in this analysis as they reported their race/ethnicity as both White and at least one REM category, and without additional information on their racial/ethnic identity, we could not easily categorize them dichotomously as White or REM. As presented in Table 2, the modified one-factor model indicated excellent fit for White/REM school counselors respectively. For White school counselors ($n = 841$), the model fit indices were as follows: $x^2(17) = 79.49$ ($p < .001$); CFI = .971; TLI = .953;

Table 2. Fit Indices for Baseline Models and Measurement Invariance Models by Race/Ethnicity.

Model	Chi-square (*df*)	CFI	TLI	RMSEA (90% CI)	SRMR
CFA baseline models (n = 1,005)					
One-factor model	209.52 (20) ***	.927	.898	.097 (.085–.109)	.038
Modified one-factor model	86.57 (17) ***	.973	.956	.064 (.051–.077)	.027
Single-group CFAs for White/REM school counselors with modified one-factor model					
White (n = 841)	79.49 (17) ***	.971	.953	.066 (.052–.081)	.030
REM (n = 144)	18.44 (17)	.996	.994	.024 (.000–.081)	.024
Multi-group CFA for White/REM school counselors with modified one-factor model					
Configural	68.50 (50)*	.983	.975	.051 (.010–.079)	.041
Metric	80.23 (57)*	.979	.973	.053 (.021–.079)	.074
Scalar	94.88 (64)**	.972	.968	.058 (.031–.082)	.078

Note. *df* = degree of freedom; CFI = comparative fit index; TLI = Tucker-Lewis index; RMSEA = root mean square error of approximation; 90% CI = 90% confidence interval; SRMR = standardized root mean square residual.

Table 3. Standardized Factor Loadings Across the SPOS-8 Models.

Model	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8
One-factor model (*n* = 1,005)	.71 (.03)	.74 (.02)	.77 (.02)	.72 (.03)	.86 (.01)	.73 (.03)	.87 (.01)	.77 (.02)
Modified one-factor model (*n* = 1,005)	.69 (.03)	.74 (.02)	.77 (.02)	.74 (.03)	.89 (.01)	.71 (.03)	.87 (.01)	.74 (.02)
Modified one-factor model for White school counselors (*n* = 841)	.68 (.03)	.74 (.02)	.77 (.02)	.73 (.03)	.88 (.01)	.68 (.03)	.87 (.02)	.75 (.03)
Modified one-factor model for REM school counselors (*n* = 144)	.69 (.08)	.80 (.05)	.75 (.06)	.77 (.08)	.89 (.03)	.81 (.05)	.90 (.02)	.70 (.06)

Note. The values in the parentheses for each item indicate standard errors. All *p* values were significant at the .001 level.

RMSEA = .066 [90% CI: .052–.081]; and SRMR = .030. For REM school counselors ($n = 144$), the model indicated a better fit: $x^2(17)$ = 18.44 (p > .05); CFI = .996; TLI = .994; RMSEA = .024 [90% CI: .000–.081]; and SRMR = .024. The absolute values of standardized factor loadings ranged from .68 to .88 for White school counselors and from .69 to .90 for REM school counselors. This result showed that the SPOS-8 measured the same constructs with the validated factor model regardless of school counselors' REM status.

Subsequently, a multi-group CFA was conducted to test the measurement invariance between White and REM school counselor groups. After the random selection of 144 White school counselors, we used data from a total of 288 school counselors for analysis. Configural, metric invariant, and scalar invariant models were fitted and compared, as presented in Table 2.

As the baseline model, the configural model with no constraints revealed CFI = .983; TLI = .975; RMSEA = .051 [90% CI: .010–.079]; and SRMR = .041), with a significant chi-square statistic [$\chi^2(50)$ = 68.50, p < .05]. This finding supports the one-factor structure of the SPOS-8 across school counselor groups by race/ethnicity. The standardized factor loadings across the four SPOS-8 models are shown in Table 3.

Next, regarding metric invariance, the model with constrained factor loadings and freely estimated intercepts indicated no statistically significant ($\triangle x^2(7)$ = 11.57, p > .05) deterioration in model fit when compared to the configural model, while three out of the four indices indicated no substantial deterioration in model fit ($\triangle$CFI = –.004, $\triangle$TLI = –.002, $\triangle$RMSEA = .002). The change in SRMR values, however ($\triangle$SRMR = .033), did exceed the criteria for invariance (Chen, 2007).

Regarding scalar invariance, the model with constrained factor loadings and intercepts differed significantly from the metric invariant model ($\triangle x^2(7)$ = 14.90, p < .05), but observed differences reflected by other model fit indices ($\triangle$CFI = –.007, $\triangle$TLI = –.005, $\triangle$RMSEA = .005, $\triangle$SRMR = .004) were not substantial (Chen, 2007). This means that White/REM school counselor groups do not vary in terms of item intercepts. Thus, strong measurement invariance for the SPOS-8 model on the current data was in evidence.

Construct Validity

Table 4 displays the descriptive statistics and correlations among the SPOS-8 total score, total and subscale scores of the CBI and SCSES, and the single items of global job satisfaction and

Table 4. Correlations of Measures with SPOS-8 Total Score, Internal Consistency Reliability, and Descriptive Statistics.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. SPOS-8 total score	(.92)														
2. CBI total score	−.50	(.90)													
3. CBI exhaustion	−.29	.82	(.89)												
4. CBI incompetence	−.27	.70	.42	(.74)											
5. CBI NWE	−.72	.75	.48	.42	(.83)										
6. CBI devaluing client	−.12	.43	.15	.40	.21	(.66)									
7. CBI DPL	−.32	.82	.70	.40	.48	.22	(.80)								
8. SCSE total score	.25	−.35	−.18	−.49	−.23	−.27	−.18	(.96)							
9. SCSE personal & social	.18	−.33	−.17	−.42	−.18	−.35	−.17	.92	(.90)						
10. SCSE L & A	.26	−.33	−.17	−.43	−.26	−.24	−.17	.90	.76	(.90)					
11. SCSE C & A	.11	−.25	−.13	−.38	−.12	−.19	−.12	.86	.74	.72	(.86)				
12. SCSE collaboration	.27	−.33	−.18	−.44	−.21	−.27	−.16	.89	.75	.74	.70	(.84)			
13. SCSE CA	.07	−.14	−.02	−.30	−.03	−.19	−.03	.70	.63	.55	.52	.60	(.71)		
14. Global job satisfaction	.60	−.58	−.38	−.43	−.62	−.23	−.38	.34	.27	.31	.22	.34	.17	–	
15. Principal support	.73	−.50	−.29	−.25	−.73	−.10	−.29	.22	.18	.24	.11	.21	.06	.59	–
M	43.46	44.69	12.10	8.84	9.72	5.18	8.76	178.92	51.36	34.2	28.56	48.07	16.72	3.97	3.68
SD	10.13	10.41	3.54	2.54	3.28	1.58	3.11	20.72	5.88	5.94	4.20	5.06	2.24	0.90	1.20

Note. $N = 1{,}005$. r = the correlation coefficient with the SPOS-8 total score. All correlation coefficients are significant at $p < .05$. The coefficient alphas (α) of global life satisfaction and school principal support were not reported since each of them has one item. SPOS-8 = Survey of Perceived Organizational Support; CBI = Counselor Burnout Inventory; SCSE = School Counselor Self-Efficacy Scale; NWE = Negative Work Environment; DPL = Deterioration in Personal Life; L & A = Leadership & Assessment; C &A = Career & Academic; CA = Cultural Acceptance; CBI and SCSE *M* and *SD* values are presented on the item scale level.

satisfaction with school principal support in our sample of 1,005 school counselors. Given the convergent correlations, our results support the construct validity of the SPOS-8 with school counselors. For instance, the SPOS-8 score correlated −.50 with the CBI Total score and .24 with the SCSE total score. Correlation coefficients ranged from −.12 to −.72 between the SPOS-8 and the CBI subscales, and ranged from .07 to .27 between the SPOS-8 and the SCSE subscales. The SPOS-8 showed a high negative correlation with the CBI Negative Work Environment subscale ($r = -.72$), along with high positive correlations with the self-perceived global job satisfaction ($r = .60$) and satisfaction with school principal support ($r = .73$).

Discussion

The Survey of Perceived Organizational Support (SPOS) is a measure designed to assess employees' perceptions of support from their workplaces. Since its development more than 30 years ago, this instrument has been employed in a variety of professions and populations, demonstrating the critical role of organizational support in multiple occupational outcomes (Rhoades et al., 2001; Riggle et al., 2009). Our study is the first attempt to date to use the SPOS-8 with a nationwide sample of professional school counselors and examine its psychometric properties with these professionals, including descriptive statistics, reliability, and validity.

Descriptive statistics obtained in this study indicate that school counselors reported a relatively high level of perceived organizational support. Particularly, the item "the school values my contribution to its well-being" represented the highest mean score. Given that supportive school settings have a significant impact on school personnel (Bardhoshi et al., 2014; Bardhoshi & Um, 2021; Yildirim, 2008), it is promising that school counselors reported high levels of perceived organizational support in the present study. For school administrators and counseling supervisors who are interested in supporting school counselors' professional performance and development, the result can be a reference not only to assess school settings, but also to identify the need for organizational changes and optimal support for school counselors' professional development. Supplementary efforts to explore the specific mechanism of organizational support (e.g., supports from administrators, peers, and students) would be helpful to clarify what is needed for supporting not only school counselors but also employees in human service professions.

In terms of evaluating internal reliability, we used the coefficient alpha recommendation of equal to or above .80 for screening purposes (Erford, 2021). Applying this interpretive guideline, the internal consistency of SPOS-8 for our sample was robust and excellent.

Regarding structural validity, this study supported the one-factor structure of SPOS-8 with a school counselor sample. Although CFA results consistently demonstrated the structural validity of a unidimensional SPOS-8 across diverse samples of human services professionals, our study is the first to endorse the single factor structure of the SPOS 8-item version with a school counseling sample. Based on the CFA results, we however identified a modified one-factor baseline model including three residual covariances, which fit our sample data. Results of single-group CFA showed that the factor structure of SPOS-8 was equivalent across White/REM school counselor groups. Measurement invariance models using multi-group CFA did not indicate any considerable differences in factor loadings and item intercepts across groups, especially considering that CFI value is regarded as a primary indicator of model invariance due to the sample-size sensitivity of RMSEA and SRMR (Chen, 2007).

In order to identify the convergent and discriminant validity of the SPOS-8, this study conducted comparisons between school counselors' perceived organizational support and other measures of relevant concepts that influence school counselors' work experiences, such as self-efficacy and burnout. Within the context of school counselors' professional experiences, the comparisons help build a deeper understanding of the SPOS-8 compared with other instruments, including the Counselor Burnout Inventory and its subscales, the School Counselor Self-Efficacy Scale and its subscales, as well as job satisfaction and satisfaction with school principal support.

As specified by Lipsey and Wilson (2001), Pearson *r*s were classified as a small (.1), medium (.3), and large (.5) effects. Among the measures, school principal support revealed the largest correlation with the SPOS-8, echoing previous studies (Hutchison, 1997a, 1997b; Rhoades et al., 2001). This result supports the initial findings of Kottke and Sharafinski (1988) that direct support from supervisors was distinguished from organizational support, although both were still highly correlated. The comparison with job satisfaction showed a large effect size (r = .60) that also aligned with existing literature reporting large effect sizes (Tourangeau et al., 2010), as well as differentiated job satisfaction from perceived organizational support (Eisenberger et al., 1997).

In terms of the CBI and its subscales, the comparisons in our sample of school counselors showed a range of negative associations, from small to large effect sizes ($r = -.12$ to $-.72$). It is notable that the CBI subscale of Negative Work Environment presented the highest correlation coefficient with the SPOS-8. These scales measure the impact of the workplace on professional counselors in similar yet distinguishable aspects of school environment. On the other hand, the smallest correlation between the CBI subscale of Devaluing the Client and SPOS-8 indicates that the two constructs have low conceptual communalities. Similarly, overall small effect sizes were generated between the SPOS-8 and SCSE and its subscales. Interestingly, the SPOS-8 showed relatively larger coefficients with the SCSE subscales of Collaboration (r = .27) and Leadership and Assessment (r = .26), which are considered partial contributors of school organizational support (Barlow & Phelan, 2007; Halawah, 2005). Consequently, understanding the perceived organizational support of school counselors in relation to personal and environmental factors is useful for developing and tailoring interventions to improve school counselor wellness.

In sum, findings with our school counselor sample study support the strong internal reliability and structural and construct validity of SPOS-8 scores, lending confidence to school counseling researchers and practitioners who are interested in a brief and accurate tool for assessing the construct of organizational support. This strong psychometric evidence, coupled with the fact that it is available free of charge, make the SPOS-8 a practical and useful way of measuring organizational support in human service professionals, including school counselors.

Limitations

This study has a few limitations to note. In terms of multicultural application, some results of this study need to be carefully interpreted since the majority of participants self-identified as White. Although we tested measurement invariance models in regard to racial/ethnic minority status, there is a possibility that our data underestimate the organizational risks that each individual ethnic minority group experiences, due to the limited number of participants for each race/ethnicity. We encourage future researchers testing the measurement invariance across diverse racial/ethnic groups to ensure sufficient participants (at least 100) for each group. Given the volunteer-based data collection used for this study, we were not able to include responses from school counselors who did not participate in the survey, which is another potential limitation. In addition, our data were collected from a single source (i.e., ASCA online membership directory), which limited the representation of school counselors without ASCA membership in our sample and has an impact on the generalizability of our findings.

Next, it should be noted that our study included cross-sectional data. While this allowed us to measure the correlations between related constructs, we were not able to determine any potential causality between them. Future studies are expected to facilitate a deeper understanding of how organizational support and related variables are influenced by employment using longitudinal results. Moreover, given that data were collected using self-report measures, it is possible that our results may be affected by the social desirability bias, especially given the sensitive nature of the construct of perceived organizational support. Moreover, because we used single-item measures, it is possible that responses reflected optimistic overestimation of job satisfaction and school principal support (Oshagbemi, 1999). Although participant responses for this online study reflected 10.5% of the original sample, a higher response rate would have contributed to more accurate measurements, and this is a limitation worth noting. Thus, caution in interpretation is recommended because there may be a sampling bias.

Finally, although we identified the applicability of the SPOS-8 to a sample of school counselors, psychometric results are always sample dependent. Thus, it is possible that the SPOS may perform differently across subsamples. Thus, further investigation of school counselors' perceptions of organizational support with diverse samples of counseling professionals, and reporting of psychometric data across samples and demographic groups can provide significant insights on this construct for the school counseling profession and encourage future comparative analyses across groups and counseling specialties.

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No potential conflict of interest was reported by the authors.

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